RATING METHODOLOGY

15 September 2025

TABLE OF CONTENTS

Scope 1
Sector overview 1
Rating approach 2
Scorecard 3
Discussion of the scorecard factors 7
Other considerations 12
Scorecard-indicated outcome 15
Assigning issuer-level and instrument-level ratings 16
Key rating assumptions 16
Limitations 17
Moody's related publications 18

Contacts

Ola Hannoun-Costa +1.212.553.1456
Associate Managing Director
ola.hannouncosta@moodys.com

Victoria Maisuradze +33.1.5330.5974
Associate Managing Director
victoria.maisuradze@moodys.com

Mariko Semetko +81.3.5408.4209
Associate Managing Director
mariko.semetko@moodys.com

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Rating Methodology

Pharmaceuticals

This rating methodology replaces the *Pharmaceuticals* methodology published on 12 November 2021. We have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Leverage and Coverage factor, we have replaced the CFO/Debt subfactor with RCF/Net Debt and replaced the Pharmaceuticals Cash Coverage of Debt subfactor with EBIT/Interest Expense. We have changed some factor weights and scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. We have also changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in the sale of pharmaceutical products applied in human healthcare. Some companies that we rate using this methodology derive revenue from other lines of business.

Companies that we rate using methodology include: (i) branded drug companies, which are engaged in the research and development of novel, patent-protected medicines; (ii) generic drug companies, which offer products that have gone off-patent by the branded pharmaceutical industry; and (iii) hybrid companies that offer a blend of branded and generic products.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Sector overview

Pharmaceuticals typically require a prescription from a physician or healthcare provider, and thereby are different from typical consumer healthcare products that are sold over the counter. Pharmaceutical companies operate under regulatory oversight in many areas, including research and clinical trials, product approval, product labeling, manufacturing standards, and product safety.

Some pharmaceutical companies maintain sizeable cash balances well in excess of operating cash needs. In some cases, companies maintain these large cash holdings to prudently manage risks from potential litigation. Additionally, pipeline investments and volatile capital

investments, as well as M&A activity, may lead to meaningful cash holdings that exceed operating cash needs.

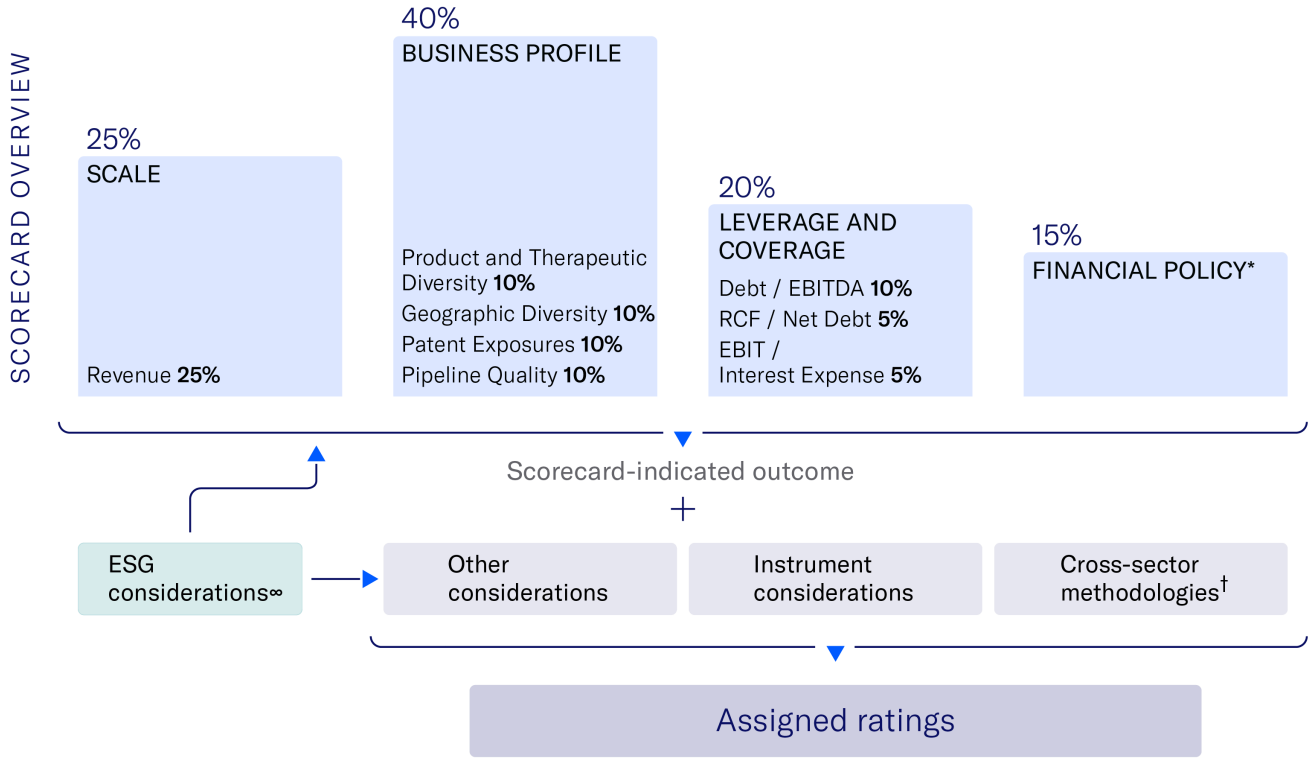
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the pharmaceuticals rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Pharmaceuticals scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (25%)									
Revenue (USD Billion) [1]	25%	≥ $60	$30 - $60	$15 - $30	$8 - $15	$3 - $8	$1 - $3	$0.25 - $1	< $0.25
Factor: Business Profile (40%)									
Product and Therapeutic Diversity	10%	**Branded Companies**: Extremely well diversified across segments, therapeutic categories and products. For example, the top 3 products are usually not expected to drive more than 25% of revenue. **Generic Companies**: Extremely well diversified across business lines, products and delivery modes. For example, over 50% of operating earnings may be derived from segments other than generic drugs.	**Branded Companies**: Very well diversified across segments, therapeutic categories and products. For example, the top 3 products are usually not expected to drive more than 30% of revenue. **Generic Companies**: Very well diversified across business lines, products and delivery modes. For example, over 40% of operating earnings may be derived from segments other than generic drugs.	**Branded Companies**: Well diversified across segments, therapeutic categories and products. For example, the top 3 products are usually not expected to drive more than 40% of revenue. **Generic Companies**: Well diversified across business lines, products and delivery modes. For example, over 30% of operating earnings may be derived from segments other than generic drugs.	**Branded Companies**: Moderately diversified across segments, therapeutic categories and products. For example, the top 3 products are usually not expected to drive more than 50% of revenue. **Generic Companies**: Moderately diversified across business lines, products and delivery modes. For example, over 20% of operating earnings may be derived from segments other than generic drugs.	**Branded Companies**: Somewhat weakly diversified across segments, therapeutic categories and products. For example, the top 3 products are usually expected to drive more than 50% of revenue. **Generic Companies**: Somewhat weakly diversified across business lines, products and delivery modes. For example, less than 20% of earnings may be derived from segments other than generic drugs; or pure-play generic company very well diversified across products and delivery modes.	**Branded Companies**: Weakly diversified across segments, therapeutic categories and products. For example, the top 3 products are usually expected to drive more than 60% of revenue. **Generic Companies**: Weakly diversified across business lines, products and delivery modes. For example, less than 10% of operating earnings may be derived from segments other than generic drugs; or pure-play generic company moderately diversified across products and delivery modes.	**Branded Companies**: Very weakly diversified across segments, therapeutic categories or products. For example, the top 3 products are usually expected to drive more than 70% of revenue. **Generic Companies**: Very weakly diversified across business lines, products and delivery modes. For example, pure-play generic company with limited diversity across products or delivery modes.	**Branded Companies**: Lack of diversification. For example, substantially reliant upon only one product. **Generic Companies**: Lack of diversification. For example, pure-play generic company with weak diversity across products or delivery modes.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Geographic Diversity	10%	Extremely well diversified geographically. For example, strong presence in at least 3 different major regions, as well as in a number of emerging markets, with the largest single region usually not expected to exceed 40% of total revenue.	Very well diversified geographically. For example, strong presence in at least 3 different major regions, as well as in a number of emerging markets, with the largest single region usually not expected to exceed 45% of revenue.	Well diversified geographically. For example, strong presence in at least 3 different major regions, as well as in a number of emerging markets, with the largest single region usually not expected to exceed 55% of revenue.	Moderately diversified geographically. For example, strong presence in at least 2 regions, with the largest single region usually not expected to exceed 65% of revenue.	Somewhat concentrated geographically. For example, may have concentration of 65%-75% of revenue in one region, with good diversity outside of this region.	Regionally very concentrated. For example, concentration of 75%-90% of revenue in one region.	Regionally extremely concentrated. For example may have > 90% of revenue in one region if the region has a diverse payor mix.	Minimal geographic diversification. For example, may have > 90% of revenue in one region with a limited number of payors.
Patent Exposures	10%	**Branded Companies**: Extremely low exposure to patent expirations and challenges. For example, revenue exposure usually not expected to exceed 10% over the next 3 years. **Generic Companies**: Extremely high barriers to entry to competing generic threats, usually resulting in expected gross margins > 80%.	**Branded Companies:** Very low exposure to patent expirations and challenges. For example, revenue exposure usually not expected to exceed 15% over the next 3 years. **Generic Companies**: Very high barriers to entry to competing generic threats, usually resulting in expected gross margins > 70%.	**Branded Companies**: Low exposure to patent expirations and challenges. For example, revenue exposure usually not expected to exceed 20% over the next 3 years. **Generic Companies**: High barriers to entry to competing generic threats, usually resulting in expected gross margins > 60%.	**Branded Companies**: Moderate exposure to patent expirations and challenges. For example, revenue exposure usually not expected to exceed 25% over the next 3 years. **Generic Companies**: Moderate barriers to entry to competing generic threats, usually resulting in expected gross margins > 50%.	**Branded Companies**: Moderately high exposure to patent expirations and challenges. For example, revenue exposure usually not expected to exceed 30% over the next 3 years. **Generic Companies**: Somewhat low barriers to entry to competing generic threats, usually resulting in expected gross margins > 40%.	**Branded Companies**: High exposure to patent expirations and challenges. For example, revenue exposure usually likely to exceed 30% over the next 3 years. **Generic Companies**: Low barriers to entry to competing generic threats, usually resulting in expected gross margins < 40%.	**Branded Companies**: Very high exposure to patent expirations and challenges. For example, revenue exposure usually likely to exceed 40% over the next 3 years. **Generic Companies**: Very low barriers to entry to competing generic threats, usually resulting in expected gross margins < 30%.	**Branded Companies**: Extremely high exposure to patent expirations and challenges. For example, revenue exposure usually likely to exceed 50% over the next 3 years. **Generic Companies**: Highly commodity-like portfolio of products, usually resulting in expected gross margins < 20%.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Pipeline Quality [2]	10%	**Branded Companies**: Excellent pipeline quality, innovation and diversity. For example, peak pipeline sales usually greater than 30% of current revenue base. **Generic Companies**: Excellent pipeline. For example, may have >= 250 ANDAs or similar filings and a very strong focus on biosimilars.	**Branded Companies**: Strong pipeline quality, innovation and diversity. For example, peak pipeline sales usually greater than 25% of current revenue base. **Generic Companies**: Strong pipeline. For example, may have 200-249 ANDAs or similar filings and a good focus on biosimilars.	**Branded Companies**: Very good pipeline quality, innovation and diversity. For example, peak pipeline sales usually greater than 20% of current revenue base. **Generic Companies**: Very good pipeline. For example, may have 150-199 ANDAs or similar filings and a focus on biosimilars.	**Branded Companies**: Moderately good pipeline quality, innovation and diversity. For example, peak pipeline sales usually greater than 15% of current revenue base. **Generic Companies**: Moderately good pipeline. For example, may have 100-149 ANDAs or similar filings and may be pursuing biosimilars, but with uncertain market position.	**Branded Companies**: Somewhat limited pipeline quality, innovation or diversity. For example, peak pipeline sales usually less than 15% of current revenue base. **Generic Companies**: Somewhat limited pipeline. For example, may have 60-99 ANDAs or similar filings and/or a lack of biosimilars.	**Branded Companies**: Limited pipeline quality, innovation or diversity. For example, peak pipeline sales usually less than 10% of current revenue base. **Generic Companies**: Limited pipeline. For example, may have 30-59 ANDAs or similar filings and/or a lack of biosimilars.	**Branded Companies**: Weak pipeline quality, innovation or diversity. For example, peak pipeline sales usually less than 5% of current revenue base. **Generic Companies**: Weak pipeline. For example, may have 15-29 ANDAs or similar filings and/or a lack of biosimilars.	**Branded Companies**: Essentially no internal late-stage pipeline; may rely solely on external R&D. **Generic Companies**: Very weak pipeline. For example may have < 15 ANDAs or similar filings with a lack of biosimilars.
Factor: Leverage and Coverage (20%)									
Debt / EBITDA [3]	10%	≤ 0.5x	0.5 - 1.5x	1.5 - 2.5x	2.5 - 3.5x	3.5 - 4.5x	4.5 - 6x	6 - 9x	> 9x
RCF / Net Debt [4]	5%	≥ 70%	50 - 70%	35 - 50%	20 - 35%	12.5 - 20%	5 - 12.5%	0 - 5%	< 0%
EBIT / Interest Expense [5]	5%	≥ 18x	12x - 18x	7x - 12x	4x - 7x	2.25x - 4x	1x - 2.25x	0.5x - 1x	< 0.5x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] ANDAs are Abbreviated New Drug Applications filed in the US. For companies with a material US presence, we typically focus solely on ANDA filings, because filings in other jurisdictions generally overlap with these US filings.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 19.5.
[5] For the linear scoring scale, the Aaa endpoint value is 20x. A value of 20x or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[1] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (25% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and a company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (40% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

A pharmaceutical company's business profile provides an important indication of the company's strength based on several measures of diversification, its exposure to patent expirations and other forms of competition, and its ability to replenish declining revenue with opportunities from its drug pipeline.

Product and Therapeutic Diversity

Business model diversity at the product and therapeutic level is important because it helps reduce a firm's exposure to numerous risks that can cause a product's revenue to erode. For branded companies, such risks include patent expirations, product safety issues, the introduction of a competitor's product or a manufacturing disruption. Operating across many different therapeutic categories provides an extra dimension of diversity, since multiple products within a therapeutic category can be affected if a competitor introduces a new product.

For generic drug companies, diversity is important but has a different meaning. Most generic companies are well diversified at the product level, but generic drugs are commodity-like, so product diversity itself may not lead to lower business risk. Other dimensions of business diversity are more important. For example, generic companies may diversify by offering less traditional product types, such as liquids, inhalers, injectables and ointments, in addition to traditional products sold as pills and capsules, which are the most

commoditized. In addition, generic companies operating other business lines are less affected by pricing pressure or other risks in their generic segment. Several generic companies have branded drug businesses, distribution businesses and active pharmaceutical ingredients businesses, all of which help reduce concentration in the pure-play generic drug business.

Geographic Diversity

Geographic diversity helps mitigate the potential reduction in profitability that can arise from exposure to regulatory or other issues that are more stringent in one region than in another. Business risks that vary by geographic region include payor negotiating leverage, pricing and budgetary pressures, healthcare reform, hurdles for regulatory approvals, workforce restructuring constraints, competition, and prevalence of litigation risks. In geographic markets with a single payor (such as a governmental entity), pricing pressure can result because the payor has authority to reduce prices. In contrast, markets with numerous payors generally afford pharmaceutical firms more pricing power than single payor markets. The single versus numerous payor distinction is critical for companies highly concentrated in one market, since concentration in a single payor market is much riskier than concentration in a market with numerous payors.

Patent Exposures

For a branded pharmaceutical company, long-lived patents are a strong barrier to entry. Patent expirations can lead to a sudden and often dramatic falloff in revenue once generic competition enters the market. The impact on profits and cash flow is often much more pronounced than the impact on revenue because products near the end of their life cycles may have very high gross margins after many years of price increases. In addition, it is difficult to scale back R&D costs and selling, general and administrative (SG&A) expenses on a basis proportional to the revenue reduction. Meanwhile, if successful, patent challenges from generic companies can cause a substantial loss of revenue years before the scheduled patent expiration.

For a generic pharmaceutical company, barriers to entry are also important because they diminish the potential for competition to erode profits. The concept of patent expiration does not readily distinguish one company from another, since the revenue base consists of products that have already lost patent protection. However, generic companies face revenue and margin declines from share loss and price erosion when other generic companies launch similar versions of their products. Generic companies that offer products that are more difficult to replicate or that have higher barriers to entry, which may include extended-release products, hormonal contraceptives and injectable products, can buffer themselves from these risks.

Pipeline Quality

The ability to offset the revenue lost to patent expiration by successfully commercializing products from the pipeline is critical to the long-term viability of branded pharmaceutical companies. A strong pipeline generating a string of blockbusters can establish a company's leadership in a therapeutic category like diabetes or oncology. Companies with strong pipelines are more likely to attract top scientists. Further, the combination of high patent exposures and a modest or a weak pipeline may also be a precursor to large-scale acquisitions for branded drug companies.

Although generic drug companies do not engage in the same type of R&D as branded drug companies, the product pipeline remains a critical indicator of future sales. The number of submitted product applications serves as an important indicator of the quality of the generic pipeline. A strong pipeline with many pending product applications shows strong competencies in chemistry and drug formulation skills, access to raw ingredients and manufacturing compliance standards. Rather than performing clinical trials to demonstrate that their products are safe and effective in treating diseases, generic companies need only demonstrate that their products are equivalent to the referenced branded product. A generic drug company' competency in challenging the patents of branded pharmaceutical companies is important to expanding its product line. Such opportunities can be lucrative for generic drug companies because regulatory statutes can extend periods of generic marketing exclusivity before other generic entrants are allowed. As patents expire on blockbuster biotech drugs, some generic companies will benefit from the production of biosimilar products. However, developing biosimilar products is much more complex than developing traditional products and requires significant capabilities in product development and manufacturing, which not all generic companies possess.

How we assess it for the scorecard

We score this factor based on four qualitative subfactors: Product and Therapeutic Diversity; Geographic Diversity; Patent Exposures; and Pipeline Quality.

Product and Therapeutic Diversity

For branded companies, we assess diversity primarily based on reported or estimated revenue breakdowns by products, therapeutic areas and business segments. We use multiple dimensions in assessing product and therapeutic diversity for branded companies. Our key considerations include but are not limited to the following:

» The proportion of total company revenue derived from the company's three largest products.

» The number of different therapeutic categories in which the company operates. For instance, two companies might have a similar portion of their revenue from their top three products, but we would view one with products spanning different therapeutic categories as more diverse than one with all three products in the same therapeutic category.

» In assessing product and segment diversity, we consider the size and profitability from business units outside of pharmaceuticals, such as consumer products, animal health and medical devices. A high proportion of revenue from these non-pharmaceutical businesses, especially if there is diversity within these businesses, significantly enhances a company's diversity at the product and therapeutic level.

For generic companies, while we consider diversity at the product level, it is typically strong, as explained above, and usually less important in our assessment. Also, it is often not disclosed by companies. Therefore, in our diversity assessment we consider several additional dimensions, such as diversity across drug delivery platforms. For instance, a company primarily offering tablets and capsules is less diverse than a company offering tablets, capsules, inhalers, injectables, ointments and patches. In addition, we consider whether the company operates in any segments other than generic pharmaceuticals, and the degree to which such segments enhance overall profitability, typically based on segment reporting.

Geographic Diversity

Our forward-looking assessment of geographic diversity is typically based on geographic revenue breakdowns provided in annual and quarterly reports. Well-diversified drug companies are not overly reliant on one geographic market. Although not all companies classify regions in the same manner in their financial reporting, most companies report their revenue breakdown across these regions: the US, Europe, Japan and emerging markets. We consider the portion of a company's sales derived from its largest geographic region as well as its presence in emerging markets. Drug companies with strong global diversification have a good sales presence in such regions, which are faster-growing than other markets.

We also consider the sales breakdown beyond just the company's top market. Two companies could have the same percentage of sales coming from their top region, but one company could operate in numerous other regions or markets, and we would typically consider it the more diverse of the two while the other might have concentrated exposure to one additional region or market and we would typically consider it less diverse. Diversity within a region is also important. If Europe represents 60% of revenue for two companies, one could derive 50% of its revenue in France, while the other could be more broadly diversified across Europe. In particular, a firm with high revenue concentration in a single region or market that generates a small percentage of the world's GDP is likely to be scored as less diverse than a firm with high revenue concentration in a developed market.

In addition, some geographic markets have single payors while others, including the US, have numerous payors. Companies with significant geographic concentration in a key market will score less favorably if this concentration is in a single-payor market because of higher exposure to pricing pressure created by payor leverage.

Patent Exposures

For branded companies, our assessment is based on the degree to which a company faces a loss of revenue or cash flow from upcoming patent exposures. We estimate the percentage of current revenue or cash flow (generally based on a last twelve month

(LTM) period) likely to be lost due to patent expirations and patent challenges over the next three years. Beyond three years we consider this risk qualitatively. Our estimates may include the following considerations:

» Revenue declines after patent expirations vary by geographic region. For patent exposures in the US market, we generally estimate that 100% of the revenue will be lost. For revenue in other countries, we generally estimate a lower percentage of revenue based on the typical erosion curves observed in those markets, which vary based on country and common local market practice.

» For products with a known upcoming patent expiration but where we estimate that generic companies will be delayed due to technical or other reasons, analysts estimate the percentage of revenue likely to be lost, evaluated on a case-by-case basis.

» For products facing pre-patent expiration challenges by generic companies, we will also include a percentage of the revenue, typically ranging from 10% to 50%, to reflect an estimated likelihood that the generic company could be successful, evaluated on a case-by-case basis.

» For biotech products facing upcoming patent expirations, we typically expect slower post-patent erosion and thus apply a percentage of revenue, typically ranging from 25% to 50%, based on our assessment of the likely timing of biosimilars and the potential impact on revenue. This reflects our view that there will generally be fewer biosimilar competitors compared to traditional generic products, and a lower degree of price discounting, resulting in less rapid erosion of revenue.

For generic companies, we primarily use a forward-looking estimate of gross margin, informed by LTM financials, as a proxy for the level of barriers to entry facing the product portfolio. Generic drug companies with very high gross margins are likely offering harder-to-manufacture products and are likely to face fewer competitors. Meanwhile, generic drug companies with low margins are likely offering more commodity-like products and are not well protected from competitive threats or pricing pressure.

For both branded and generic companies, we also consider any unique situations involving abnormally high pricing pressure on key products, which can arise if payors demand higher rebates or if government payors implement large price cuts. The resulting price reduction can impact a company's financial performance in a manner similar to a product rolling off patent, and can therefore affect our assessment of this subfactor.

For companies with both branded and generic business segments, we consider the patent exposures and barriers to entry within each to arrive at an overall blended score.

Pipeline Quality

For branded companies, in our assessment of pipeline quality we consider pipeline diversity and innovation. Our assessment of pipeline quality is primarily based on our estimate of peak sales from pipeline compounds in late-stage development, and we may also consider the degree of product risk. We consider peak pipeline sales relative to the current revenue base, typically using LTM total revenue as the reference point.

In arriving at our estimates, we typically consider the size of the therapeutic category, the level of unmet medical need and the competitive environment, among other factors. For branded drug companies, our estimates may be further informed by the uncertainties related to timing of approval and launch, pricing, labeling (narrow versus broad safety warnings), the impact of related products going generic, and the impact of future market entrants from other companies' pipelines. We often compare and contrast our estimates to those from other market participants when available. We may adjust peak sales for a given product upward or downward over time as new data emerges, or for other reasons, including competitive dynamics. We primarily consider only those products in late-stage development (in the US, Phase III clinical trials or later) because probability of success for earlier stage products is less certain and the development time frame is longer. Other factors that we typically consider in our assessment include the following:

» The degree of approval risk. For instance, we generally consider pipeline drugs high-risk opportunities when many companies have unsuccessfully attempted Phase III trials (Alzheimer's disease treatments have been a notable example). As such, we may adjust peak sales for certain products downward to reflect a lower probability of success.

- The degree of product diversity in the pipeline. For instance, two companies could have the same aggregate peak sales estimate, but for one company it substantially relates to one key pipeline drug, while for another company it is spread across many different drugs. The latter company would likely score higher in this subfactor.
- We consider line extensions of existing products on a case-by-case basis, since line extensions frequently do not lead to significant incremental sales, and we usually haircut the incremental sales estimate by up to 75%. The rationale for this reduction includes the difficulty in estimating incremental sales, the potential that some off-label use could already be occurring, and our view that new chemical compounds add more strength to the pipeline.
- For companies with substantial operations outside of branded pharmaceuticals (which may include consumer products, medical devices, diagnostics or animal health, among others), we do not explicitly project peak sales for the product launches that could arise from these business segments, since many of them could cannibalize existing product lines. However, a company's level of innovation in its other business segments can favorably impact its score for this subfactor.

Generic companies rarely disclose the full details regarding which products are in their pipelines. Instead, we typically evaluate the number of disclosed regulatory filings and first-to-file opportunities, as well as the potential success of any patent challenges. For companies that have a material presence in the US, this assessment is primarily based on a count of Abbreviated New Drug Application (ANDA) filings. ANDA filings are production applications awaiting regulatory review and are specific to the US generic pharmaceuticals market. For companies with a material presence in the US, we typically focus solely on ANDA filings, because filings in other jurisdictions generally overlap with these US filings.

In non-US markets, companies follow other regulatory approval procedures (which may be similar to the ANDA process). To assess pipelines for generic companies that do not have a material presence in the US generic market, we use available equivalent data sources to approximate the number of unique compounds in the pipeline that have been submitted for regulatory approval. For generic companies, we also consider the extent to which the pipeline involves the development of biosimilar products.

For companies with both branded and generic business segments, we consider the pipeline scores within each to arrive at an overall blended score.

Factor: Leverage and Coverage (20% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: Debt/EBITDA; RCF/Net Debt; and EBIT/Interest Expense.[2]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand business lines, consolidate market positions, advance cost synergies or seek access to new areas of R&D.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[3]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely

short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[4]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's geographic diversity and pipeline quality. However, we may assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[5]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[6]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[7]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[8]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[9]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[10] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

© 2025 Moody's Corporation, Moody's Investors Service, Inc., Moody's Analytics, Inc. and/or their licensors and affiliates (collectively, "MOODY'S"). All rights reserved.
CREDIT RATINGS ISSUED BY MOODY'S CREDIT RATINGS AFFILIATES ARE THEIR CURRENT OPINIONS OF THE RELATIVE FUTURE CREDIT RISK OF ENTITIES, CREDIT COMMITMENTS, OR DEBT OR DEBT-LIKE SECURITIES, AND MATERIALS, PRODUCTS, SERVICES AND INFORMATION PUBLISHED OR OTHERWISE MADE AVAILABLE BY MOODY'S (COLLECTIVELY, "MATERIALS") MAY INCLUDE SUCH CURRENT OPINIONS. MOODY'S DEFINES CREDIT RISK AS THE RISK THAT AN ENTITY MAY NOT MEET ITS CONTRACTUAL FINANCIAL OBLIGATIONS AS THEY COME DUE AND ANY ESTIMATED FINANCIAL LOSS IN THE EVENT OF DEFAULT OR IMPAIRMENT. SEE APPLICABLE MOODY'S RATING SYMBOLS AND DEFINITIONS PUBLICATION FOR INFORMATION ON THE TYPES OF CONTRACTUAL FINANCIAL OBLIGATIONS ADDRESSED BY MOODY'S CREDIT RATINGS. CREDIT RATINGS DO NOT ADDRESS ANY OTHER RISK, INCLUDING BUT NOT LIMITED TO: LIQUIDITY RISK, MARKET VALUE RISK, OR PRICE VOLATILITY. CREDIT RATINGS, NON-CREDIT ASSESSMENTS ("ASSESSMENTS"), AND OTHER OPINIONS INCLUDED IN MOODY'S MATERIALS ARE NOT STATEMENTS OF CURRENT OR HISTORICAL FACT. MOODY'S MATERIALS MAY ALSO INCLUDE QUANTITATIVE MODEL-BASED ESTIMATES OF CREDIT RISK AND RELATED OPINIONS OR COMMENTARY PUBLISHED BY MOODY'S ANALYTICS, INC. AND/OR ITS AFFILIATES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT CONSTITUTE OR PROVIDE INVESTMENT OR FINANCIAL ADVICE, AND MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT AND DO NOT PROVIDE RECOMMENDATIONS TO PURCHASE, SELL, OR HOLD PARTICULAR SECURITIES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT COMMENT ON THE SUITABILITY OF AN INVESTMENT FOR ANY PARTICULAR INVESTOR. MOODY'S ISSUES ITS CREDIT RATINGS, ASSESSMENTS AND OTHER OPINIONS AND PUBLISHES OR OTHERWISE MAKES AVAILABLE ITS MATERIALS WITH THE EXPECTATION AND UNDERSTANDING THAT EACH INVESTOR WILL, WITH DUE CARE, MAKE ITS OWN STUDY AND EVALUATION OF EACH SECURITY THAT IS UNDER CONSIDERATION FOR PURCHASE, HOLDING, OR SALE.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS, AND MATERIALS ARE NOT INTENDED FOR USE BY RETAIL INVESTORS AND IT WOULD BE RECKLESS AND INAPPROPRIATE FOR RETAIL INVESTORS TO USE MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS OR MATERIALS WHEN MAKING AN INVESTMENT DECISION. IF IN DOUBT YOU SHOULD CONTACT YOUR FINANCIAL OR OTHER PROFESSIONAL ADVISER.
ALL INFORMATION CONTAINED HEREIN IS PROTECTED BY LAW, INCLUDING BUT NOT LIMITED TO, COPYRIGHT LAW, AND NONE OF SUCH INFORMATION MAY BE COPIED OR OTHERWISE REPRODUCED, REPACKAGED, FURTHER TRANSMITTED, TRANSFERRED, DISSEMINATED, REDISTRIBUTED OR RESOLD, OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE, IN WHOLE OR IN PART, IN ANY FORM OR MANNER OR BY ANY MEANS WHATSOEVER, BY ANY PERSON WITHOUT MOODY'S PRIOR WRITTEN CONSENT. FOR CLARITY, NO INFORMATION CONTAINED HEREIN MAY BE USED TO DEVELOP, IMPROVE, TRAIN OR RETRAIN ANY SOFTWARE PROGRAM OR DATABASE, INCLUDING, BUT NOT LIMITED TO, FOR ANY ARTIFICIAL INTELLIGENCE, MACHINE LEARNING OR NATURAL LANGUAGE PROCESSING SOFTWARE, ALGORITHM, METHODOLOGY AND/OR MODEL.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT INTENDED FOR USE BY ANY PERSON AS A BENCHMARK AS THAT TERM IS DEFINED FOR REGULATORY PURPOSES AND MUST NOT BE USED IN ANY WAY THAT COULD RESULT IN THEM BEING CONSIDERED A BENCHMARK.
All information contained herein is obtained by MOODY'S from sources believed by it to be accurate and reliable. Because of the possibility of human or mechanical error as well as other factors, however, all information contained herein is provided "AS IS" without warranty of any kind. MOODY'S adopts all necessary measures so that the information it uses in assigning a credit rating is of sufficient quality and from sources MOODY'S considers to be reliable including, when appropriate, independent third-party sources. However, MOODY'S is not an auditor and cannot in every instance independently verify or validate information received in the credit rating process or in preparing its Materials.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability to any person or entity for any indirect, special, consequential, or incidental losses or damages whatsoever arising from or in connection with the information contained herein or the use of or inability to use any such information, even if MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers is advised in advance of the possibility of such losses or damages, including but not limited to: (a) any loss of present or prospective profits or (b) any loss or damage arising where the relevant financial instrument is not the subject of a particular credit rating assigned by MOODY'S.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability for any direct or compensatory losses or damages caused to any person or entity, including but not limited to by any negligence (but excluding fraud, willful misconduct or any other type of liability that, for the avoidance of doubt, by law cannot be excluded) on the part of, or any contingency within or beyond the control of, MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers, arising from or in connection with the information contained herein or the use of or inability to use any such information.
NO WARRANTY, EXPRESS OR IMPLIED, AS TO THE ACCURACY, TIMELINESS, COMPLETENESS, MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE OF ANY CREDIT RATING, ASSESSMENT, OTHER OPINION OR INFORMATION IS GIVEN OR MADE BY MOODY'S IN ANY FORM OR MANNER WHATSOEVER.
Moody's Investors Service, Inc., a wholly-owned credit rating agency subsidiary of Moody's Corporation ("MCO"), hereby discloses that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by Moody's Investors Service, Inc. have, prior to assignment of any credit rating, agreed to pay Moody's Investors Service, Inc. for credit ratings opinions and services rendered by it. MCO and all MCO entities that issue ratings under the "Moody's Ratings" brand name ("Moody's Ratings"), also maintain policies and procedures to address the independence of Moody's Ratings' credit ratings and credit rating processes. Information regarding certain affiliations that may exist between directors of MCO and rated entities, and between entities who hold credit ratings from Moody's Investors Service, Inc. and have also publicly reported to the SEC an ownership interest in MCO of more than 5%, is posted annually at ir.moodys.com under the heading "Investor Relations — Corporate Governance — Charter and Governance Documents - Director and Shareholder Affiliation Policy."
Moody's SF Japan K.K., Moody's Local AR Agente de Calificación de Riesgo S.A., Moody's Local BR Agência de Classificação de Risco LTDA, Moody's Local MX S.A. de C.V, I.C.V., Moody's Local PE Clasificadora de Riesgo S.A., and Moody's Local PA Calificadora de Riesgo S.A. (collectively, the "Moody's Non-NRSRO CRAs") are all indirectly wholly-owned credit rating agency subsidiaries of MCO. None of the Moody's Non-NRSRO CRAs is a Nationally Recognized Statistical Rating Organization.
Additional terms for Australia only: Any publication into Australia of this document is pursuant to the Australian Financial Services License of MOODY'S affiliate, Moody's Investors Service Pty Limited ABN 61 003 399 657AFSL 336969 and/or Moody's Analytics Australia Pty Ltd ABN 94 105 136 972 AFSL 383569 (as applicable). This document is intended to be provided only to "wholesale clients" within the meaning of section 761G of the Corporations Act 2001. By continuing to access this document from within Australia, you represent to MOODY'S that you are, or are accessing the document as a representative of, a "wholesale client" and that neither you nor the entity you represent will directly or indirectly disseminate this document or its contents to "retail clients" within the meaning of section 761G of the Corporations Act 2001. MOODY'S credit rating is an opinion as to the creditworthiness of a debt obligation of the issuer, not on the equity securities of the issuer or any form of security that is available to retail investors.
Additional terms for India only: Moody's credit ratings, Assessments, other opinions and Materials are not intended to be and shall not be relied upon or used by any users located in India in relation to securities listed or proposed to be listed on Indian stock exchanges.
Additional terms with respect to Second Party Opinions and Net Zero Assessments (as defined in Moody's Ratings Rating Symbols and Definitions): Please note that neither a Second Party Opinion ("SPO") nor a Net Zero Assessment ("NZA") is a "credit rating". The issuance of SPOs and NZAs is not a regulated activity in many jurisdictions, including Singapore. JAPAN: In Japan, development and provision of SPOs and NZAs fall under the category of "Ancillary Businesses", not "Credit Rating Business", and are not subject to the regulations applicable to "Credit Rating Business" under the Financial Instruments and Exchange Act of Japan and its relevant regulation. PRC: Any SPO: (1) does not constitute a PRC Green Bond Assessment as defined under any relevant PRC laws or regulations; (2) cannot be included in any registration statement, offering circular, prospectus or any other documents submitted to the PRC regulatory authorities or otherwise used to satisfy any PRC regulatory disclosure requirement; and (3) cannot be used within the PRC for any regulatory purpose or for any other purpose which is not permitted under relevant PRC laws or regulations. For the purposes of this disclaimer, "PRC" refers to the mainland of the People's Republic of China, excluding Hong Kong, Macau and Taiwan.

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the issuer/deal page on https://ratings.moodys.com for the most updated credit rating action information and rating history.

REPORT NUMBER 1439384

Author

Dirk Steinicke

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454